Academic year 2025-2026 (Even Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**MAY 2026**	**Maximum Marks**	**50**
Course Code	**HS361TA**	**Duration**	**90 Min**
Sem	**VI Semester**	**Test-2**	
Entrepreneurship and Intellectual Property Rights			

PART – A

S No	Questions	M	BT	CO
1	What is the primary objective of registering an Industrial Design?	02	L2	5
2	Copy Right protects expression but not the ideas. Justify?	02	L2	5
3	Define Performers Rights.	02	L1	5
4	Define 'Organic Growth' in the context of business expansion.	02	L2	1
5	What is a Break-Even Point in the context of business operations?	02	L2	3

PART – B

S No	Questions	M	BT	CO
1.	A company plans to launch an electric scooter in a semi-urban market where fuel prices are rising, but charging infrastructure is limited. Explain how the company should assess market feasibility and demand before launching the product.	10	L3	2
2.	A startup is launching an AI-based online tutoring platform connecting freelance tutors with students across India. Explain what key elements should be included in the executive summary of its business plan.	10	L4	3
3.	A food startup has developed a unique recipe and process for manufacturing a long-lasting organic juice. The company wants to prevent competitors from copying it. Explain how the firm can protect its trade secrets in India and why such protection is significant.	10	L3	5
4.	An educator creates a comprehensive online course including recorded lectures, PDFs, animations, and quizzes, and uploads it on a learning platform. Later, another platform copies the videos and distributes them without permission. Explain the nature, scope, and rights violated in this situation.	10	L3	5
5.	A startup proposes an AI-based mobile app that detects plant diseases using images. Explain how the startup should evaluate technical feasibility through prototype development and proof of concept before full-scale launch.	10	L3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max	Quiz	06	04				04	06				
	Marks	Test	10	10	20	10			30	10	10		

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th Dec 2025	**CIE - II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: ENTREPRENEURSHIP & INTELLECTUAL PROPERTY RIGHTS		**Course Code : HS351TA**

Part A

Sl. No	Questions	M	BT	CO
1.	Distinguish between Proof of Concept (PoC) and Prototype with respect to risk mitigation.	2	1	3
2.	Why is technical feasibility meaningless without market feasibility? Justify.	2	1	3
3.	How does reverse engineering weaken trade secret protection?	2	2	3
4.	Why is novelty mandatory but aesthetic appeal critical in industrial design protection?	2	3	3
5.	Define Copyright.	2	2	3

Part B

Sl. No	Questions	M	BT	CO
1.	Explain the elements of a business plan. Describe each component with suitable examples.	10	2	2
2.	Discuss the complete Entrepreneurial Opportunity Evaluation Process covering: • Identification of market opportunities and trends • Technical feasibility (Prototype and Proof of Concept) • Financial feasibility (Cost estimation, revenue projection, and break-even analysis) • Role of cross-disciplinary collaboration in technological innovation	10	3	3
3.	Explain the acts which do not constitute Infringement under the Copyright Law.	10	2	3
4.	Explain in detail the complete procedure for obtaining Industrial Design protection in India. Further, critically evaluate: • Grounds and procedure for revocation of registered designs • Types of design infringement • Civil and criminal remedies available Illustrate your answer with at least one real case study.	10	3	2
5.	A software company develops a unique algorithm for data security and keeps it confidential. A former employee illegally discloses this information to a competitor. 1. Identify the type of intellectual property right involved. 2. Explain the legal remedies available under Indian law. 3. Discuss the importance of trade secret protection with reference to this case.	10	3	3

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 6th Apr 2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Entrepreneurship & Intellectual Property Rights		**Course Code : HS361TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define Eco-Labelling.	2	2	4
2.	Differentiate between trademark and passing off. (Any two points)	2	2	5
3.	What is a Collective Trademark?	2	2	4
4.	What is a Break-even point in the context of business operations?	2	2	3
5.	Why is a financial plan important in a business plan?	2	2	3
	Part – B			
1.	Discuss any two important trademark infringement cases, highlighting their impact on trademark law.	10	3	3
2.	Explain the difference between registrable and non-registrable trademarks. Discuss the criteria that make a trademark registrable and provide examples of marks that cannot be registered.	10	3	4
3a.	The Soil Association, a UK-based organic certification organization, owned a registered eco-label that certified products meeting strict organic farming standards. The Association of British Growers (ABG) used a mark similar to the Soil Association's eco-label on their products without permission. Whether ABG's unauthorized use of the Soil Association's eco-label constituted trademark infringement and whether it misled consumers into believing the products were certified organic by the Soil Association. Discuss.	05	3	5
3b.	ITC Ltd., an Indian conglomerate, owned the trademark "Munch" for its chocolate products. Nestle India Ltd. started marketing a chocolate bar also named "Munch." ITC alleged that Nestle's use of the mark was deceptively similar and infringed its trademark rights. • Whether Nestlé's use of the mark "Munch" infringed ITC's registered trademark by creating deceptive similarity. • Whether customers were likely to be confused or misled by the similarity of the marks. • Is Nestlé's use amounted to unfair competition or passing off.	05	4	5

4.	Discuss the following aspects related to new product development and innovation: a) How can identifying market opportunities and trends influence the ideation process? b) Explain the role of integration of engineering principles and cross-disciplinary collaboration in technological innovation. c) Describe the methods to assess market feasibility and demand analysis for a new product. d) Discuss how technical feasibility is evaluated through prototype development and proof of concept. e) Outline the key components of financial feasibility analysis including cost estimation and revenue projection.	10	2	3
5.	Describe the SWOC analysis and explain Porter's generic competitive strategies with examples.	10	3	3

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Go, change the world®

USN

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**07/01/2025**	**Maximum Marks**	**10 + 50**
Course Code	**HS351TA**	**Duration**	**120 Min**
Sem	**V**	**CIE -II**	
ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	_____ component is key in assessing market feasibility	1	1	2
2	What does break-even analysis determine?	1	1	3
3	Which strategy in Porter's generic strategies focuses on offering unique products?	1	1	3
4	What does a proof of concept primarily demonstrate?	1	1	2
5	Why is prototyping important in the ideation process?	2	1	2
6	In order to register Industrial Design, Design should be new and _____	1	1	4
7	Can a colour by itself be subject matter of registration of industrial design? If yes, justify your statement.	2	2	4
8	Upon expiry of registration of a design under Designs Act 2000 it becomes _____ property	1	1	4
	Part –B			
1	A medium-sized technology company in India develops an innovative algorithm for optimizing supply chain logistics. To maintain its competitive edge, the company decides not to patent the algorithm but to treat it as a trade secret. Over time, a former employee joins a competitor and allegedly discloses key aspects of the algorithm. The company must navigate the legal landscape to protect its trade secrets. **Questions:** i. What legal tools and mechanisms are available in India to protect trade secrets? ii. Analyze the company's decision to treat the algorithm as a trade secret instead of patenting it. What are the advantages and risks of this approach? What preventive measures could the company have implemented to minimize the risk of trade secret leakage?	10	3	4

2	A furniture design firm creates a unique chair design that combines aesthetics with ergonomic principles. The design becomes a market hit, but a competitor begins manufacturing and selling a chair with an almost identical appearance. The firm initiates legal proceedings to obtain design protection and stop the competitor's activities. Questions: i. What features of the chair design qualify it for industrial design protection?	10	3	4

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15/12/2025	**CIE – II-A**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Operations Management		**Course Code : IM352IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	In time series models, regular fluctuations occurring within a year due to weather or festivals are called _____ variations.	1	1	1
2.	The difference between the actual value and the forecasted value is known as the _____.	1	1	2
3.	Qualitative forecasting methods rely heavily on _____ judgment.	1	1	1
4.	A hotel predicts weekend occupancy using past booking patterns. This uses a _____ time series model.	1	2	1
5.	A company uses a senior manager's opinion to forecast sales of a new product. This reflects the _____ forecasting method.	1	1	1
6.	A production line experiencing a delay at one critical machine identifies it as the _____ resource.	1	2	4
7.	A production manager selects the job with the shortest processing time to reduce average waiting time. This applies the _____ rule.	1	2	3
8.	A plant with multiple machines generates several two-machine equivalent sequences to approximate the optimal schedule. This procedure follows the _____.	1	2	3
9.	A manufacturer with two machines creates an optimal job sequence by choosing the smallest processing time among both machines. This applies _____ rule.	1	2	3
10.	For two jobs processed on multiple machines, the job with smaller processing time on the first machine is scheduled _____ in the Johnson-like rule for job shops.	1	2	3
	Part – B			
1	Explain the importance of measuring forecast error in Operations Management. Discuss the managerial implications of each measure with suitable examples.	08	2	3
2	With suitable example, differentiate between forward scheduling and backward scheduling	08	2	4

<table>
<tr><td>3</td><td>A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as 10, 11, 10, 11, 14, 15, 14 & 15 units respectively. What would be his estimate of sales for September?
(i) If he has followed 4 month weighted moving average (with weights 0.4, 0.3, 0.2 & 0.1).
(ii) If he has followed SES with $\alpha = 0.2$ and his estimated sales for January is 11.</td><td>12</td><td>4</td><td>3</td></tr>
<tr><td>4</td><td>Consider the following single machine scheduling problem.
<table>
<tr><th>Job, j</th><th>Processing time, t_j</th><th>Due date, d_j</th></tr>
<tr><td>1</td><td>12</td><td>17</td></tr>
<tr><td>2</td><td>10</td><td>14</td></tr>
<tr><td>3</td><td>10</td><td>12</td></tr>
<tr><td>4</td><td>9</td><td>20</td></tr>
<tr><td>5</td><td>14</td><td>18</td></tr>
<tr><td>6</td><td>17</td><td>27</td></tr>
</table>
Determine the sequence which will minimize the maximum lateness (L_{max}). Also, determine the L_{max} with respect to the optimal sequence.</td><td>10</td><td>4</td><td>4</td></tr>
<tr><td>5</td><td>Use graphical method to minimize the time needed to process the following jobs on the machines shown (i.e. for each machine, find the job which should be scheduled first). Also, calculate the total time elapsed to complete both jobs.
<table>
<tr><td rowspan="2">Job 1</td><td>Sequence</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr>
<tr><td>Time (Hrs).</td><td>3</td><td>4</td><td>2</td><td>6</td><td>2</td></tr>
<tr><td rowspan="2">Job 2</td><td>Sequence</td><td>B</td><td>C</td><td>A</td><td>D</td><td>E</td></tr>
<tr><td>Time (Hrs)</td><td>5</td><td>4</td><td>3</td><td>2</td><td>6</td></tr>
</table></td><td>12</td><td>4</td><td>4</td></tr>
</table>

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

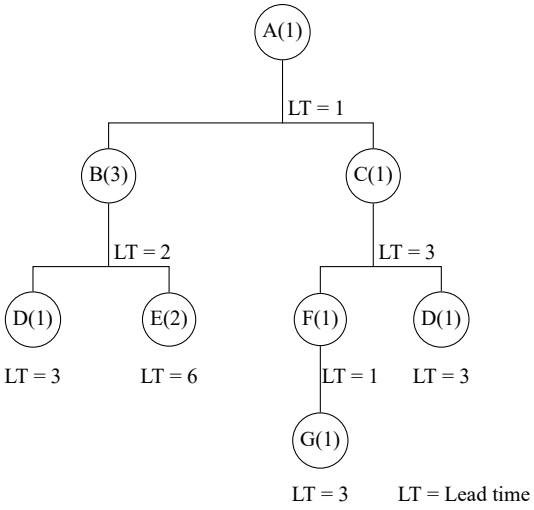
Date	**07 January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM352IA**	**Duration**	**120 Min**
Sem	**V**	**CIE – II**	
OPERATIONS MANAGEMENT			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	In services, the process structure is primarily influenced by the level of _____ and customization.	1	1	2
2	In manufacturing, a _____ process is ideal for high-volume, standardized production.	1	2	2
3	The first step in a systematic approach to long-term capacity decisions is _____ future demand.	1	2	2
4	A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____.	1	1	2
5	The _____ component of a forecast captures recurring patterns over specific periods, such as seasons or months.	1	2	3
6	The smoothing constant in exponential smoothing ranges between 0 and _____.	1	1	3
7	____ provides a detailed plan of what needs to be produced and when, serving as an input for MRP.	1	1	3
8	The ____ file contains information about the quantities of materials required to make a product.	1	2	3
9	_____ integrates all business processes, including finance, HR, and supply chain, into a unified system.	1	2	3
10	_____ ensures that operational capabilities are in line with the company's strategic goals.	1	2	3
	Part –B			
1	Explain the major process strategy decisions and their implications for operations.	10	2	2

Academic year 2024-2025 (ODD Sem)

2	Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices. **a.** If other factors are equal, which two offices should be located closest together? **b.** Figure shows an alternative layout for the department. What is the total weighted-distance score for this plan based on rectilinear distance and assuming that offices A and B are three units of distance apart? **CLOSENESS MATRIX** Trips Between Departments Department: A — 25; B — —; C — 90; D — —; E — 105; F — 125 Department: A — —; B — 125; C — 25; D — 105; E — —; F — — **C F A** **B E D**	10	3	2
3	The monthly demand for units manufactured by the Acme Rocket Company has been as follows: May 100; June 80; July 110; August 115; September 105; October 110; November 125; December 120 **i.** Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; $\alpha = 0.2$. **ii.** Calculate the absolute percentage error for each month from June through December and the MAD and MAPE of forecast error as of the end of December. **iii.** Calculate the tracking signal as of the end of December. What can you say about the performance of your forecasting method?	10	3	3

4	Refer to the bill of materials for product A. If there is no existing inventory and no scheduled receipts, how many units of items G, E, and D must be purchased to produce five units of end item A?  	10	4	3
5	Explain how the concept of dependent demand in material requirements planning is fundamental to resource planning.	10	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	–	4	6	–	4	6	–	–	–	–
	Test		–	20	30	–	–	20	20	10	–	–

****

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 16th Dec 2025	**CIE - II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: FINANCIAL ACCOUNTING & COSTING		**Course Code : IM254TA**

Note: Answer all the Questions.

Sl. No	Questions	M	BT	CO
	PART - A			
1.1	What is meant by Overheads in Costing?	2	1	2
1.2	Differentiate between Fixed Costs and Variable Costs.	2	3	2
1.3	Identify two sectors for which activity-based costing is suitable.	2	3	4
1.4	Comment on how cost accumulation happens in Process Costing with an example.	2	3	4
1.5	Job costing is for _____, _____ jobs, while batch costing is for *a group of* _____ *products* treated as one batch.	2	2	3
	PART - B			
2.a	Enumerate the objectives of Cost Accounting.	05	2	3
2.b	Discuss the various elements of Costing.	05	2	3
3	Prepare Trial Balance for the following Balances as on 31st March 2023	10	3	2

Balances	**₹**
Cost of Goods Sold	5,20,000
Opening Stock	50,000
Closing Stock	50,000
Salary and Wages	50,000
Sales	8,00,000
Plant & Machinery	2,00,000
Drawing	50,000
Investment	4,30,000
Creditors	1,00,000
Capital	4,00,000

4.	Prepare Profit and Loss account from the following particulars for the year ended March 31, 2023: Gross Profit ₹ 4,69,000 Subletting ₹ 10,000 Commission Received ₹ 40,000 Rent, tax, and Insurance ₹ 10,000 Discount Received ₹ 22,000 Export Duty ₹ 15,000 Rent Received ₹ 23,000 Audit Fees ₹ 5,000 Bank Charges₹ 8,000 Packing Charges(Show Room)₹ 12,000 Dividend Received ₹ 15,000 Depreciation ₹ 24,000 Administration Expenses ₹ 35,000 Selling Expenses ₹ 25,000 Printing and Stationery ₹ 15,000 Distribution Expenses ₹ 28,000 Office Expenses ₹ 32,000 Trade Expenses ₹ 50,000 General Expenses ₹ 25,000 Carriage outward ₹ 15,000 Legal Expenses ₹ 5,000 Bad Debts ₹ 10,000	10	4	3
5	Write short notes on any three types of costing methods used in organizations.	10	2	3
6	In a manufacturing unit, raw material passes through four processes, I, II, III, and IV and the output of each process is the input for the subsequent process. The losses in the four processes are respectively 25%, 20%, 20% and 16 2/3 % respectively for I, II, III and IV processes of the input. If the end product at the end of the IV process is 40,000 kg, what is the quantity of raw material required to be fed at the beginning of Process I and the cost of the same at Rs. 5 per kg?	10	3	3

*****

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**8th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM254TA**	**Duration**	**120 Mins**
Sem	**V Semester**	**CIE – II**	
FINANCIAL ACCOUNTING AND COSTING			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	What is a Trial Balance, and why is it prepared?	02	2	CO2
2	How is Gross Profit calculated?	02	1	CO3
3	Why is depreciation included in the Income Statement?	02	2	CO3
4	What is the basic accounting equation represented in a Balance Sheet?	02	1	CO3
5	What is the difference between current and non-current assets?	02	1	CO1

PART – B

Sl. No	Questions	M	BT	CO
1.	The following are the ledger balances of a business as of December 31, 2024. Prepare a trail balance.	10	3	CO2

Account	Amount (₹)	Type
Cash	50,000	Debit
Capital	1,50,000	Credit
Sales	2,00,000	Credit
Purchases	1,00,000	Debit
Furniture	25,000	Debit
Salaries	30,000	Debit
Accounts Receivable	45,000	Debit
Accounts Payable	40,000	Credit

Academic year 2024-2025 (ODD Sem)

2.	Prepare a Profit and Loss Account for Green Enterprises for the year ending December 31, 2024, from the following details: **Particulars** / **Amount (₹)** Sales Revenue 7,50,000 Purchases 4,00,000 Opening Inventory 1,20,000 Closing Inventory 80,000 Wages 50,000 Office Expenses 40,000 Utilities 30,000 Depreciation on Machinery 20,000	10	4	CO3
3.	Prepare a Balance Sheet for Green Valley Enterprises as of December 31, 2024, from the following details: **Particulars** / **Amount (₹)** / **Particulars** / **Amount (₹)** Capital 5,00,000 / Cash 1,00,000 Bank Loan 3,00,000 / Building 5,00,000 Accounts Payable 1,50,000 / Inventory 2,00,000 Salaries Payable 50,000 / Furniture 80,000 Accounts Receivable 1,20,000 / Equipment 1,50,000	10	3	CO3
4.	From the following information, prepare a cost sheet for XYZ Manufacturing Co. for the year ending March 31, 2025: **Particulars** / **Amount (₹)** Raw Materials Consumed 3,00,000 Direct Labor 2,00,000 Factory Overheads 80,000 Office and Administrative Overheads 40,000 Selling and Distribution Overheads 30,000 Opening Stock of Finished Goods 50,000 Closing Stock of Finished Goods 60,000 Profit Percentage on Cost 25%	10	3	CO4
5.	Explain the three main elements of cost sheet with examples. How are they classified and used in cost accounting?	10	3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	02	02	06	–	–	06	04	–	–	–	–
		Test	–	10	20	20	–	–	–	30	20	–	–

*****

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 16th Dec 2025	**CIE-II**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Quality Assurance		**Course Code : IM353IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Explain the purpose of a p-chart.	2	3	01
2.	What is AOQ?	2	2	01
3.	State the difference between single and double sampling plans.	2	2	02
4.	What is AOQL?	2	3	02
5.	What are multiple sampling plans?	2	2	02
	6.			
7.	Draw the type - A OC curve for the single sampling plan: $n = 50, C = 1, N = 200$.	10	2	3
2	Differentiate between C chart and U chart	10	3	3
3	Lots of Bushing screws from a single supplier were inspected 100 percent with the following results. Lot No. / Amount in lot / Number rejected 01 / 6,000 / 183 02 / 6,000 / 131 03 / 10,000 / 184 04 / 1,783 / 23 05 / 3,089 / 171 06 / 5,774 / 65 07 / 3,000 / 46 08 / 2,724 / 85 Draw an appropriate control chart for the data. What can you conclude on the state of control?	10	4	3

<table>
<tr><td>4</td><td>The following table gives the number of missing rivets at the aircraft final inspection:
<table>
<tr><th>Sl No</th><th>Air craft No.</th><th>Number of missing rivets</th><th>Sl No</th><th>Air craft No.</th><th>Number of missing rivets</th></tr>
<tr><td>1</td><td>202</td><td>07</td><td>6</td><td>207</td><td>11</td></tr>
<tr><td>2</td><td>203</td><td>12</td><td>7</td><td>208</td><td>06</td></tr>
<tr><td>3</td><td>204</td><td>13</td><td>8</td><td>209</td><td>19</td></tr>
<tr><td>4</td><td>205</td><td>18</td><td>9</td><td>210</td><td>04</td></tr>
<tr><td>5</td><td>206</td><td>17</td><td>10</td><td>211</td><td>26</td></tr>
</table>
Establish an appropriate control chart for the purpose.</td><td>10</td><td>4</td><td>4</td></tr>
<tr><td>5</td><td>Explain the meaning of the following terms:
<table>
<tr><td>i.</td><td>Producer’s risk</td><td>ii.</td><td>AQL</td></tr>
<tr><td>iii.</td><td>Consumer’s risk</td><td>iv.</td><td>LTPD</td></tr>
</table></td><td>10</td><td>3</td><td>4</td></tr>
</table>

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**7th January 2025**	**Maximum Marks**	**50**
Course Code	**IM353IA**	**Duration**	**90 Min**
Sem	**V**	**CIE – II**	
QUALITY ASSURANCE			

Note:
1. Answer all the Questions.

PART – A **Max Marks: 10**

Sl. No	Questions	M	BT	CO
1.	Differentiate between Attribute control chart and Variable control chart.	2	L2	CO1
2.	What does an R-chart measure, and why is it important in statistical process control?	2	L1	CO2
3.	What is the key difference between a p-chart and a np-chart?	2	L2	CO3
4.	Expand the acronyms: AQL, UTPD	2	L2	CO3
5.	Write two applications of acceptance sampling.	2	L2	CO2

PART – B

Sl. No	Questions	M	BT	CO
1.	A TiW layer is deposited on a substrate using a sputtering tool. The following table contains layer thickness measurements (in Angstroms) on 10 subgroups of four substrates. Set up $\bar{X}$ and s charts on this process and compute trial control limits.	10	3	CO3
2.	Write a note on Standardized control chart. What is the process capability? Explain any one of the techniques of process capability analysis.	10	2	CO1

Subgroup	x_1	x_2	x_3	x_4
1	459	449	435	450
2	443	440	442	442
3	457	444	449	444
4	469	463	453	438
5	443	457	445	454
6	444	456	456	457
7	445	449	450	445
8	446	455	449	452
9	444	452	457	440
10	432	463	463	443

Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>3.</td><td>Lots of Bushing screws from a single supplier were inspected 100 percent with the following results.
<table>
<tr><th>Lot No.</th><th>Amount in lot</th><th>Number rejected</th></tr>
<tr><td>1</td><td>6000</td><td>183</td></tr>
<tr><td>2</td><td>6000</td><td>131</td></tr>
<tr><td>3</td><td>10000</td><td>184</td></tr>
<tr><td>4</td><td>1783</td><td>23</td></tr>
<tr><td>5</td><td>3089</td><td>171</td></tr>
<tr><td>6</td><td>5774</td><td>65</td></tr>
<tr><td>7</td><td>3000</td><td>46</td></tr>
<tr><td>8</td><td>2724</td><td>85</td></tr>
</table>
Draw an appropriate control chart for the data. What can you conclude on the state of control?</td><td>10</td><td>2</td><td>CO2</td></tr>
<tr><td>4.</td><td>Draw the OC Curve, AOQ Curve, ASN Curve and ATI Curve for the single sampling plan with N = 5000, n = 100 and c = 2</td><td>10</td><td>4</td><td>CO4</td></tr>
<tr><td>5.</td><td>A factory inspects different samples of a product, recording the number of defects in each sample. The sample size is constant, and the number of defects is provided below:
<table>
<tr><th>Sample No.</th><th>Number of Defects</th></tr>
<tr><td>1</td><td>5</td></tr>
<tr><td>2</td><td>8</td></tr>
<tr><td>3</td><td>6</td></tr>
<tr><td>4</td><td>4</td></tr>
<tr><td>5</td><td>7</td></tr>
<tr><td>6</td><td>9</td></tr>
<tr><td>7</td><td>5</td></tr>
<tr><td>8</td><td>6</td></tr>
</table>
i. Draw an appropriate control chart for the data. Evaluate the process.
ii. Plot the centreline, upper control limit (UCL), and lower control limit (LCL).</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**		CO1	CO2	CO3	CO4	L1	L2	L3	L4
	Quiz	Max Marks	02	04	04	–	2	8	–	–
	Test		10	20	10	10	–	30	10	10

****